Probabilistic Bayesian Network Classifier for Face Recognition in Video Sequences

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***Abstract*—The inherent properties of video sequences allow for representation of data in both spatial and temporal dimensions. Using conventional image-based methods for face recognition in video is often an ineffective approach as the essential spatio-temporal properties are not fully harnessed. This paper proposes a probabilistic Bayesian network classifier to accomplish effective recognition of faces in video sequences. In our model, we introduce a joint probability function that encodes the causal dependencies between video frames, selected exemplars or representative images of a video, and subject classes. This enables both the temporal continuity between video frames and also the spatial relationships between exemplars and their respective exemplar-set classes to be captured. To simplify the tedious estimation of densities, the proposed method also utilizes probabilistic similarity scores that are computationally inexpensive. Good recognition rates were achieved by our proposed method in comprehensive experiments conducted on two standard face video datasets.**

***Keywords*-Bayesian network, classifier, video-based face recognition.**

I. Introduction

Bayesian networks have been frequently used in various real-world applications such as medical diagnosis, forecasting, manufacturing processes and automated vision systems [1], [2] to great success. The robust use of Bayesian networks in these applications is primarily due to its capability in encoding dependencies among variables in a particular problem domain and learning causal relationships between these variables.

The task of learning Bayesian networks has been known as a *NP-Hard* search problem [3]. Most Bayesian networks are trained or learned from a particular problem domain with large amounts of data, using necessary heuristic techniques to overcome its computational complexity. In this work, we delegate the task of learning from training data to other efficient techniques while focusing on the design of a probabilistic Bayesian network classifier that is tailored to solve a multivariate recognition problem.

A well known example of a simple yet effective Bayesian network is the Naive Bayes (NB) classifier [4]. The NB classifier is a particular Bayesian network where every attribute has its corresponding node connected only to the node representing the class. Despite its simplicity, the NB classifier has been reported to yield good results in various practical domains and applications, and it can also be trained efficiently in a supervised learning setting. The main drawback of the NB classifier is that classification accuracy can be easily hampered due to its failure to identify causality between multiple attributes or when the attributes are not totally independent.

A. Related Work in Face Recognition

Research in automatic face recognition has grown rapidly in the past few decades, with a primary focus towards still image-based recognition. A wide variety of current state-of-art methods [5] have been reported to be effective in recognizing human faces. However, these methods tend to perform poorly in the presence of complex face variations in unconstrained environments. With the emergence of video, the abundance of data has presented sufficient motivation to formulate this recognition task as a compound decision problem [4].

A landmark psychological and neural study [6] reported that the human brain performs recognition of faces by means of both invariant structure of features and also idiosyncratic movements and gestures. Biologically, the media temporal cortex of a human brain performs motion processing, which aids the recognition of dynamic facial signatures. Inspired by these findings, the inherent temporal dynamics in sequentially-ordered video data can also be exploited to improve machine recognition of faces.

Many recent works [7], [8], [9], [10], [11], [12], [13] have been motivated to improve machine recognition of faces by exploiting temporal dynamics in video sequences. Zhou *et al.* [7] proposed a joint probability distribution of identity and head motion using sequential importance sampling (SIS) for simulataneous tracking and recognition. Lee *et al.* [8] approximated a nonlinear appearance manifold as a set of linear sub-manifolds, and transition probabilities were learned to model the connectivities between sub-manifolds. Due to their simplistic modeling of densities, these methods are highly sensitive to conditions where statistical correlations between training and test data are weak.

Methods that utilize Hidden Markov Models (HMM) to capture facial dynamics in video have also generated much interest in recent years [9], [10]. Despite showing promising results, the main disadvantage of these methods

is that the learning of temporal dynamics during recognition is computationally demanding, rendering these approaches practically infeasible.

Exemplar-based approaches have become increasingly popular in the literature, where certain works have applied statistical learning to exploit temporal dynamics. Fan *et al.* [12] proposed a Bayesian inference model by constructing view specific sub-manifolds for classification. Liu *et al.* [13] formulated a spatio-temporal embedding based on Bayesian keyframe learning and statistical classification. None of these methods consider the influence of different exemplars with respect to their respective exemplar-set classes.

In this paper, we introduce a probabilistic Bayesian network classifier for the purpose of face recognition in video sequences. On the premise of an exemplar-based recognition approach, the proposed classifier is formulated based on a joint probability function that seeks to incorporate temporal continuity between successive video frames and to encode the causal relationship between exemplars and their respective exemplar-set classes. We also present non-parametric approaches to the estimation of probability densities using similarity scores that are computationally inexpensive.

The remainder of this paper is organized as follows. The next section introduces the relevant attributes in our problem domain with a brief overview of the face recognition framework. Section 3 presents the theoretical foundation of the proposed classifier. Section 4 discusses the experiments along with its results and comparisons with other classification methods. Finally, Section 5 concludes the paper and suggests some potential future work.

II. Video-based Face Recognition Framework

Before we describe the proposed classifier in detail, the environment settings of a video-based face recognition framework are formalized to provide a better overview of the domain attributes that will be used in our model. We also briefly describe some steps taken to prepare both training and test data for classification.

A. Domain Setting

The abundance of face images in video poses a methodological challenge. While conventional still image-based face recognition is a straightforward matching of a test image to a gallery of training images, it is an ill-posed problem for video sequences. The availability of multiple image frames in a training video requires further simplification to formulate a feasible classification methodology.

Assuming that each subject has only one training video, we represent each training video by a set of *exemplars*, or representative face images that were selected from the video sequence. This setting is similar to that used in [11], [12].

For general notation, a sequence of face images extracted from a video is given as,

$$X_c = \{x_{c,1}, x_{c,2}, \ldots, x_{c,N_c}\} \ , \tag{1}$$

where N_c is the number of face images in the video.

Assuming that each video contains the faces of the same person and c is the subject identity of a C-class problem, $c \in \{1, 2, \ldots, C\}$, the associated exemplar set is denoted by

$$E_c = \{e_{c,1}, e_{c,2}, \ldots, e_{c,M}\} \ , \tag{2}$$

Thus, the overall exemplar-class set can be succinctly summarized as

$$E = \{e_{c,m} | c = 1, \ldots, C; m = 1, \ldots, M\} \ , \tag{3}$$

in which $e_{c,m} \subseteq X_c$ and there are a total of $C \times M$ unique exemplar-classes.

In cases where more than one training video of a particular class is used, image frames from all similar-class videos are aggregated before the exemplars are extracted.

B. Data Preparation

Considering the large amount of face variations in each training video sequence, we first apply the nonlinear dimensionality reduction method, Locally Linear Embedding (LLE) [14] to learn a low-dimensional manifold embedding from its original data space. Then, faces in each training video sequence are partitioned into clusters using hierarchical agglomerative clustering (HAC) [4], [15]. Following that, the face exemplar set is constructed by selecting the face images that are nearest to the mean of each cluster.

To obtain meaningful discriminative features from the face exemplar set (in Eq. 3), we apply a state-of-the-art nonlinear supervised projection method, Neighborhood Discriminative Manifold Projection (NDMP) [16], which is effective in uncovering the nonlinear data manifold of video sequences. Face images from test videos are also projected to the NDMP-space for the subsequent classification task.

III. Proposed Bayesian Network Classifier

In a Bayesian inference model, the subject identity of a test video X can be found by estimating the *maximum a posteriori* (MAP) probability decision rule,

$$c^* = arg \ \max_{C} \ P(c | x_{1,\ldots,N_c}), \tag{4}$$

where the subscript notation of x succinctly represents a sequence of N_c images.

A. Naive Bayes Classifier

In a typical Naive Bayes classifier, estimation based on MAP decision rule can be expressed as

$$P(c|x_{1,\ldots,N_c}) = \frac{P(c)P(x_{1,\ldots,N_c}|c)}{P(x_{1,\ldots,N_c})} = \frac{P(c)P(x_{1,\ldots,N_c}|c)}{\sum_c P(x_{1,\ldots,N_c}|c)P(c)} \ , \tag{5}$$

where $P(c)$ is the prior probability of each class, $P(x_{1,\ldots,N_c}|c)$ is the likelihood probability of x given class c and the denominator is a normalization factor to ensure that the sum of the likelihoods over all possible classes equals to 1. Assuming conditional independence between all observations *i.e.* $x_i \perp\!\!\!\perp x_j|c$ where $i \neq j$, Eq. (5) can be rewritten as,

$$P(c|x_{1,\ldots,N_c}) = \prod_{i=1}^{N_c} \frac{P(c)P(x_i|c)}{P(x_i)} \ . \tag{6}$$

B. Exemplar-based Bayesian Network (EBN) Classifier

We propose an Exemplar-based Bayesian Network (EBN) classifier by introducing a joint probability function,

$$P(c, E, X) = P(X|E)P(E|c)P(c) \ , \tag{7}$$

where the exemplar-class set E is a new latent variable.

Thus, the MAP classifier in Eq. 4 is redefined by maximizing the *joint posterior probability* of the class c and exemplar-class E given observation X:

$$\begin{aligned} \max_C \ P(c, E|X) &= \max_C \ \frac{P(c, E, X)}{P(X)} \\ &= \max_C \ \sum_{j=1}^{M} \frac{P(X|e_{c,j})P(e_{c,j}|c)P(c)}{P(X)} \\ &= \max_C \ \sum_{j=1}^{M} \prod_{i=1}^{N_c} \frac{P(x_i|e_{c,j})P(e_{c,j}|c)P(c)}{P(x_i)} \ . \end{aligned} \tag{8}$$

The graphical model of the EBN classifier is shown in Fig. 1.

Intuitively, the conditional probability $P(e_{c,j}|c)$ acts as an *exemplar prominence* weight for the class likelihood $P(x_i|e_{c,j})$. The marginal probability $P(x_i)$ does not depend on both c and E, thus functioning only as a normalization constant. Since the class prior probability $P(c)$ is assumed to be non-informative at the start of observation sequence X, fixing uniform priors is a reasonable estimation.

In conventional Bayesian classifiers, a multivariate Gaussian density function is used to estimate data distribution. However with the limited sample size of our problem, accurate estimation of distribution can be challenging and easily result in overfitting or underfitting of data. An alternative method of performing non-parametric density estimation is by using distance measures (or kernel density estimator with uniform kernel), which are computationally inexpensive.

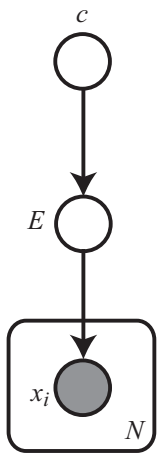


Figure 1. Graphical model of the proposed EBN classifier

The exemplar likelihood and exemplar prominence probability are determined as follows:

1) Computation of Exemplar Likelihood $P(x_i|e_{c,j})$: We define Frame Similarity Score (FSS) as the reciprocal of the ℓ^2-norm distance between the observed face image x_i and the j-th exemplar-class $e_{c,j}$,

$$S_i^{FSS}(x_i, e_{c,j}) = \frac{1}{d_{\ell^2}(x_i, e_{c,j})} \ . \tag{9}$$

The likelihood of the test face image x_i given the class c and exemplar-class e is determined by the ratio of FSS for exemplar-class $e_{c,j}$ to the total sum of FSS across all $C \times M$ exemplar-classes,

$$P(x_i|e_{c,j}) = \frac{S_i^{FSS}(x_i, e_{c,j})}{\sum_{k=1}^{C} \sum_{m=1}^{M} S_i^{FSS}(x_i, e_{k,m})} \ . \tag{10}$$

2) Computation of Exemplar Prominence $P(e_{c,j}|c)$: Causal relationship between exemplars and their parent classes is represented by the exemplar prominence probability $P(e_{c,j}|c)$. Similar to the computation of likelihood probability, we avoid estimating density functions by representing the influence of an exemplar in its own class subspace by a normalized Hausdorff distance metric [17],

$$d_h(e_{c,j}, E_c) = \frac{1}{\Lambda} \min_{e' \in E_c} \|e_{c,j} - e'\| \ , \tag{11}$$

where E_c is the exemplar set of class c and Λ is a sum normalization factor to ensure distances are relative within-class. By defining the Exemplar Prominence Score (EPS) as the reciprocal of the distance metric,

$$S_{c,j}^{EPS}(E_c, e_{c,j}) = \frac{1}{d_h(e_{c,j}, E_c)} \tag{12}$$

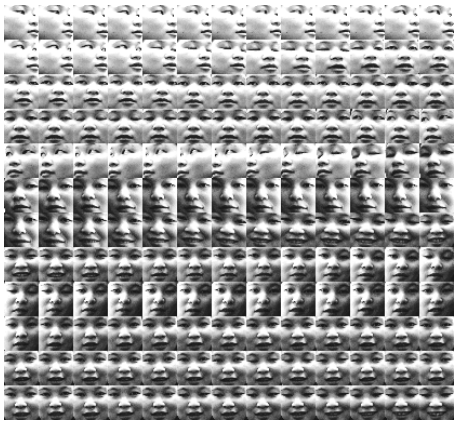
the exemplar prominence probability can be formulated as

$$P(e_{c,j}|c) = \frac{S_{c,j}^{EPS}(E_c, e_{c,j})}{\sum_{m=1}^{M} S_{c,j}^{EPS}(E_c, e_{c,m})} \ , \qquad (13)$$

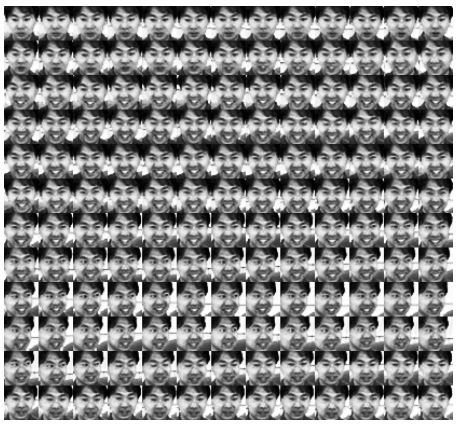
which can be pre-computed offline since it does not depend on input observation X.

IV. Experimental Results and Discussions

To evaluate our proposed classifier, we conducted experiments on two standard video face datasets – the Honda/UCSD [8] and CMU MoBo [18].



(a) Honda/UCSD



(b) CMU MoBo

Figure 2. Sequentially-ordered image frames of a sample video sequence taken from the evaluated datasets.

(a) Honda/UCSD

(b) CMU MoBo

Figure 3. Sample extracted exemplar sets (one set in each row).

The first dataset, Honda/UCSD, which was collected specifically for video-based face recognition, consists of 59 video sequences of 20 different people (each person has at least 2 videos). Each video contains about 300–600 frames, comprising of large pose and expression variations with complex out-of-plane (3-D) head rotations.

The second dataset, CMU MoBo is one of the most widely used benchmark dataset for video-based face recognition. It consists of 96 sequences of 24 different subjects. Each subject has 4 sequences captured in different walking situations: holding a ball, fast walking, slow walking, and walking on the incline. While moving, the subjects also change their facial expressions frequently. Each video contains about 300 frames.

For each video sequence, faces images are extracted using the Viola-Jones cascaded face detector [19] and resampled to 32×32 pixels. Histogram equalization was applied to normalize lighting effects. To systematically evaluate all possible subjects in each database, we use one video sequence for training and the remaining video sequences for testing. The test set is constructed by randomly sampling 20 subsequences of 100 frames each from every test video. The number of extracted exemplars is fixed at $M = 7$ for Honda/UCSD and $M = 6$ for CMU MoBo. Fig. 3 shows some sample extracted exemplar sets from both datasets. The rule of thumb for determining the value of M is by identifying the "elbow" or trade-off point of the residual error curve from the earlier HAC clustering step [4].

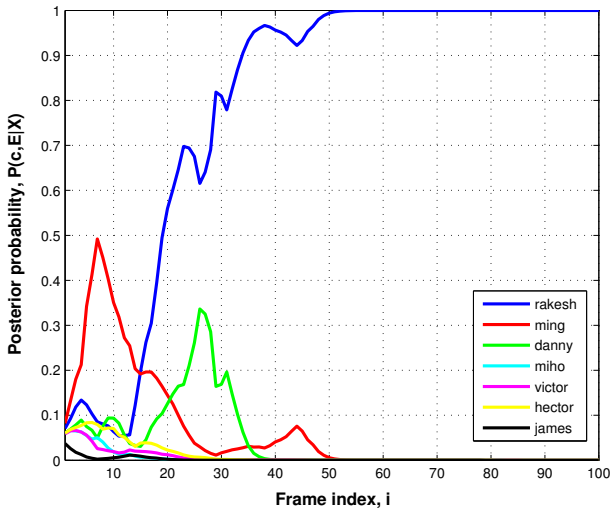
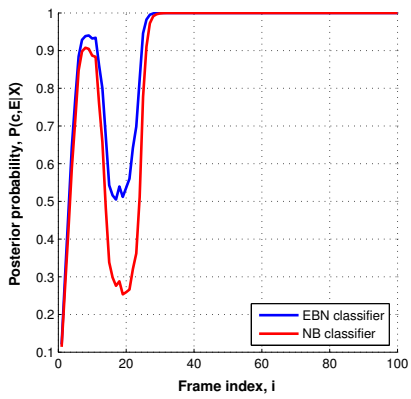


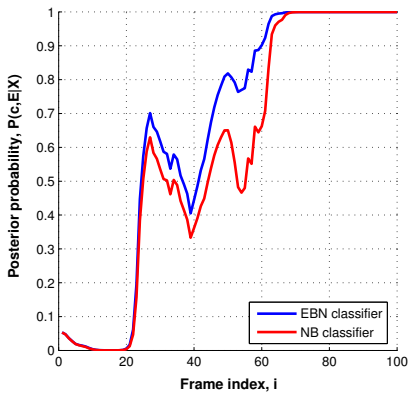
Figure 4. Posterior probability $P(c, E|X)$ versus frame index plot of the *rakesh* test video from Honda/UCSD dataset. Posterior probabilities of the seven most probable subjects are shown in different colors. The subject (in blue line) is correctly identified.

Evaluation of a test video sequence with the proposed classifier is accomplished by maximizing the MAP decision rule in Eq. (8) across all frames in a test sequence to decide on the subject identity. In Fig. 4, the posterior plot of a

sample test sequence from the Honda/UCSD dataset demonstrates that the proposed Bayesian recognition framework is capable of arriving at the correct identity, even when the initial frames were incorrectly classified. By comparing the posterior probabilities of the EBN and NB classifiers across frames on a same test sequence (in Fig. 5), we observed that the EBN classifier is able to achieve a slightly higher posterior probability during classification. It is also less susceptible to sudden drops of confidence level, when there are more doubtful faces within the sequence.



(a)



(b)

Figure 5. Comparisons between the posterior probability versus frame index plots of the EBN (blue line) and NB (red line) classifiers on the (a) *chia* video and (b) *louis* video, of the Honda/UCSD dataset.

We first compare the evaluation performance of the following classification schemes on both datasets:

1) **Majority voting** (with 1-nearest neighbor matching), where a vote is taken in each frame and the class with the majority vote is classified as the subject.
2) **Probabilistic voting**, where the likelihood probabilities of each frame (based on Eq. (10)) are combined cumulatively by simple sum rule. The class with the largest sum of likelihoods is classified as the subject.
3) **Naive Bayes classifier** (based on Eq. (5))
4) **Exemplar-based Bayesian Network classifier** (based on Eq. (8))

Results on the Honda/UCSD and CMU MoBo datasets in Table I show that our proposed EBN classifier is able to outperform other methods in an exemplar-based face recognition setting. Generally, Bayesian classifiers yield better recognition rates than traditional voting schemes in a video-based setting. Despite the rapidly changing face pose and expression within a same video sequence, the temporal continuity between frames are well-established in our statistical framework. Compared to the Naive Bayes (NB) classifier, the proposed EBN classifier further enhances recognition accuracy as the causal relationship between the exemplars and their respective exemplar-class sets were given consideration. As a result, the more prominent exemplars are given more attention and influence in the inferencing process, and vice versa.

Table I
RECOGNITION RATES (%) OF VARIOUS CLASSIFICATION METHODS ON EVALUATED DATASETS

Classification method	**Dataset**	
	Honda/UCSD	**CMU MoBo**
Majority vote	63.38	94.65
Probabilistic vote	70.78	95.08
NB classifier	84.61	96.30
EBN classifier	86.90	97.70

In our second evaluation, we tested the classification schemes using different extracted features on the CMU MoBo dataset. Table II shows the robustness of the proposed classifier as it consistently yields higher recognition rates even when different features (PCA [20], LDA [21], NPE [22]) are used.

In computational terms, the EBN classifier has a time complexity of $\mathcal{O}(nm)$ compared to $\mathcal{O}(n)$ for a typical Naive Bayes classifier, where n is the number of frames in a video sequence and m is the number of exemplars per class. The difference is negligible in our problem setting as the usage of exemplar-classes (refer to Section II-A) results in $\mathcal{O}(nm)$ for Naive Bayes as well, and under typical conditions, $m \ll n$. The only drawback of the EBN lies in its space complexity of $\mathcal{O}(cm)$ (c for number of classes) due to the offline computation of exemplar prominence values.

Table II
RECOGNITION RATES (%) WITH VARIOUS SETTINGS ON THE CMU MOBO DATASET

Classification method	**Feature**			
	PCA	**LDA**	**NPE**	**NDMP**
Majority vote	83.28	90.95	93.53	94.65
Probabilistic vote	82.16	92.93	92.33	95.08
NB classifier	85.63	93.66	93.27	96.30
EBN classifier	86.44	94.97	94.11	97.70

V. Conclusion

In this paper, we present a probabilistic Bayesian network classifier for face recognition in video, which introduces causal relationships between extracted exemplars and their respective exemplar-set classes while incorporating temporal continuity between consecutive video frames. We also simplify the density estimation by utilizing probabilistic distance measures that are computationally lightweight. By evaluation on two standard face video datasets, the proposed classifier achieved better recognition rates compared to conventional voting methods and the Naive Bayes classifier. Its consistent performance using different features also demonstrated the robustness of this classification scheme. In future work, the current model can be further extended to accommodate new probabilistic measures for image sets (linear subspaces) and manifolds. In practical, more tests can be conducted in various difficult and adversed real-world scenarios to further fine-tune the proposed method.

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